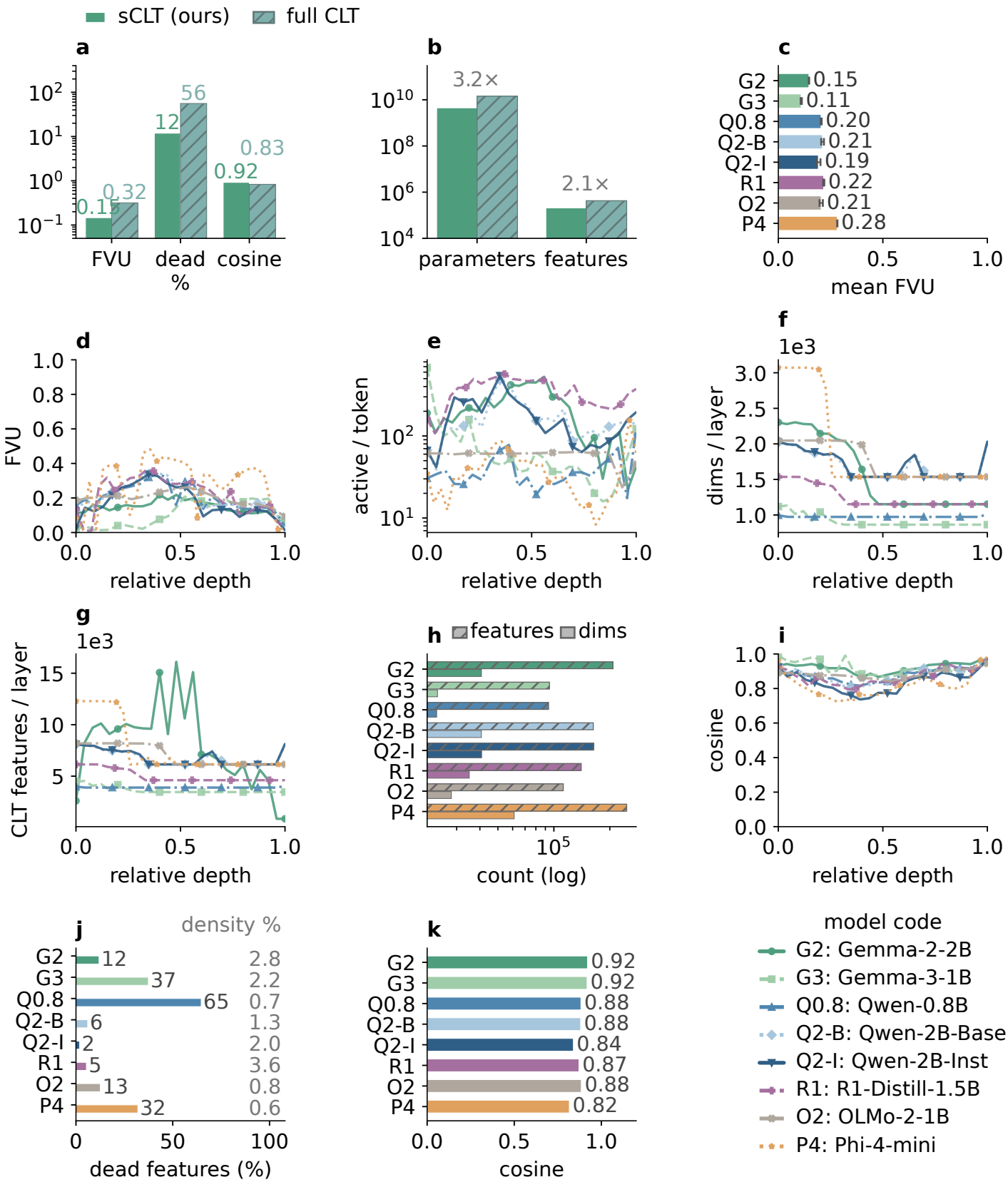
sCLT (ours)
full CLT
a
FVU
dead %
cosine
0.15
0.32
12
56
0.92
0.83
b
3.2×
2.1×
parameters
features
c
G2 0.15
G3 0.11
Q0.8 0.20
Q2-B 0.21
Q2-I 0.19
R1 0.22
O2 0.21
P4 0.28
mean FVU
d
FVU
relative depth
e
active / token
relative depth
f
1e3
dims / layer
relative depth
g
1e3
CLT features / layer
relative depth
h
features
dims
count (log)
i
cosine
relative depth
j
density %
G2 12 2.8
G3 37 2.2
Q0.8 65 0.7
Q2-B 6 1.3
Q2-I 2 2.0
R1 5 3.6
O2 13 0.8
P4 32 0.6
dead features (%)
k
G2 0.92
G3 0.92
Q0.8 0.88
Q2-B 0.88
Q2-I 0.84
R1 0.87
O2 0.88
P4 0.82
cosine
model code
G2: Gemma-2-2B
G3: Gemma-3-1B
Q0.8: Qwen-0.8B
Q2-B: Qwen-2B-Base
Q2-I: Qwen-2B-Inst
R1: R1-Distill-1.5B
O2: OLMo-2-1B
P4: Phi-4-mini